PR

Principal Component Analysis (PCA)

- Dimensionality Reduction
- Definition
  - Orthogonal projection
  - lower dimension
  - variance maximised (unsupervised)
- Steps
  - Standardise the data
  - Compute the covariance matrix
  - calculate the eigenvectors (PC) and the eigenvalues (variances of PCs)
  - select a feature vector (with largest eigenvalue)
  - project data onto principal components $\tilde{x}_n = u_1^T x_n$
- Scree plot: help to decide the number of dimensions (find an elbow)

Linear Discriminant Analysis (LDA)

- Definition
  - maximise separability (supervised)
  - numerator: used to maximise group distance (between-class scatter)
  - denominator: used to minimise data overlap (within-class scatter)
- Steps
  - Standardise the data
  - Calculate values
    - Calculate means
    - Calculate within-class & between-class scatter matrix
    - Combine scatter matrices (Fisher Criterion)
  - Finding optimasation
    - Calculate eigencectors and eigencalues
    - Sort eigenvalues and select feature vectors
  - Project the data

Clustering

- Two common methods
  - K-means Clustering
    - Steps
      - Randomly initialise k centroids
      - Assign each datapoint to the nearest centroid
      - Calculate the means of each cluster's datapoints
      - Update the centroids to the calculated means
      - Back to step 2 until values converge
    - Elbow method: help to decide k
  - Hierarchical Clustering
    - Classification
      - Agglomerative Algorithm: Bottom-up approach
      - Divisive Algorithm: Top-down approach
    - Steps (Agglomerative)
      - All points are their own clusters
      - Group the closest two cluster into a new cluster
      - Repeat the step until there is only one cluster
    - Dendrogram — Y-axis is distance between connections (similarity)
- Distance Metrics
  - Euclidean distance $d_E(p,q) = \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}$
  - Manhattan distance (taxicab) $d_M(p,q) = \sum_{i=1}^{n}|p_i - q_i|$
  - Chebyshev distance $d_C(p,q) = \max_i |p_i - q_i|$

Decision Trees

- Structure
  - node: test on the data
  - branch: result of the test
  - leaf node: final decision (predefined class)
- Growing method:
  CART (Classification and Regression Tree)
  - Classification Tree
    - Gini diversity index: $Gini(D) = 1 - \sum_{i=1}^{k} p_i^2$
      (measures the impurity of the categories)
    - Calculate the Gini index of each features
    - Choose the feature that minimises Gini index as a new node
    - Continue the calculation and division on child nodes
  - Regression Tree
    - MSE: $MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \bar{y})^2$
    - Calculate the MSE of each features
    - Choose the feature that minimises MSE as a new node
    - Continue the calculation and division on child nodes
- Problem: overfitting, poor generalisation

Ensemble learning

- Combine weak algorithms together
  - Classification: voting
  - Regression: calculating mean
- Bagging
  - Bootstrap Aggregation (data resampling)
  - Random Forests
    - Sampling method
      - Sample bagging
      - Feature bagging
    - out-of-bag (OOB) error: used to evaluate the performance / optimise the forest size
- Boosting
  - Mechanics
    - Generate weak learners sequentially
    - Each tries to improve previous result
    - Decision trees are stumps (with just a root and two leaves)
  - Adaboost
    - Use random sample of data to generate a stump
    - Update the weight of each decision tree's vote
    - Adjust the weight of the sample according to the result
    - Back to the first step

Sequential (Time Series) Data

- Dynamic Time Warping (DTW)
  - Definition
    - Warp two time series data to try to align them
    - Used for detect similarities
  - Mechanics
    - Divide time sequences into equal numbers of samples
    - Calculate the Euclidean distance between all samples
    - Use minimum distances to measure similarity
    - Use minimum distances to stretch / contract points over time (non linear)